Thermodynamic geometry of black holes

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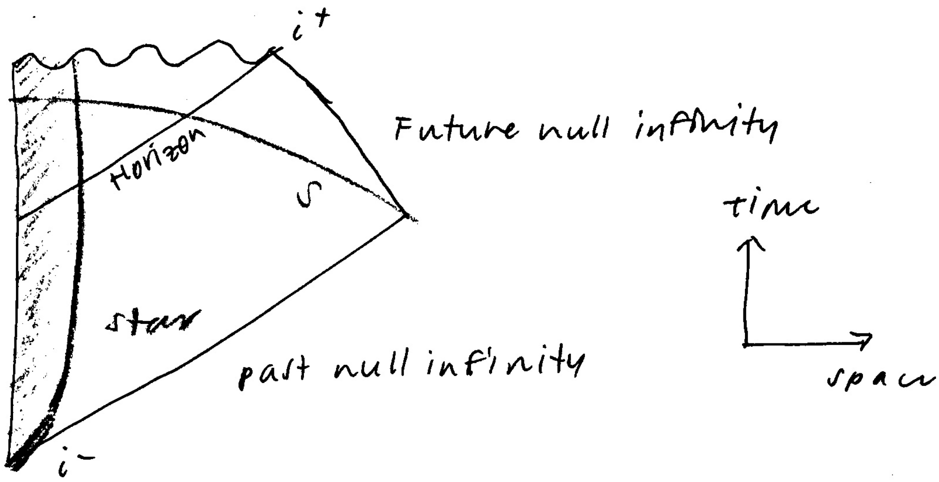
Contents

1 Black hole thermodynamics

1.1 The Hawking temperature of a BH

- Notes from Witten [1].
- Bekenstein's formula for the entropy of a BH is
$$S = \frac{A}{4G\hbar},$$
and the generalized entropy is (setting $\hbar = 1$)
$$S_{\text{gen}} = \frac{A}{4G} + S_{\text{out}},$$
where the first term is the black hole entropy, and the second term is the ordinary entropy of matter and radiation outside the horizon. (The latter needs to be replaced by von Neumann entropy when the system is not in thermal equilibrium.) This then admits a generalized second law:
$$\frac{dS_{\text{gen}}}{dt} \geq 0.$$
- However, this law appears to be problematic even for Schwarzschild absorbing a beam of black body radiation of temperature T, because then
$$\Delta S_{\text{gen}} = \left(8\pi GM - \frac{4}{3T}\right)E,$$
and this becomes negative if the typical photon wavelength (which is $O(1/T)$) is much larger than the Schwarzschild radius. The resolution is to understand that the black hole is actually strongly emitting such photons.
- As an aside: the entropy of a macroscopic black hole, which is massive ($\sim e^{A/4G}$), might seem to be in tension with the no-hair theorem. But even an ordinary thermal system exhibits a sort of no-hair behavior. Pour a cup of water into a glass. It will slosh around for a while, obviously not in thermal equilibrium. But after some time, the transients will die and the water will be in apparent thermal equilibrium, although its detailed microscopic state will not be describable by a truly thermal density matrix.
- We discuss the resolution of this decreasing-entropy paradox. Witten gives an argument in the spirit of Hawking's original work, but with a modification adapted from Fredenhagen and Haag which involves tracing back to initial conditions on a conveniently chosen hypersurface S that intersects the horizon outside the worldvolume of the collapsing star. This evades any discussion of physics at high energies.
- A very rough summary (for this is quite technical): take a coordinate function u on S, which vanishes on the horizon and is positive outside it. Then one can write
$$t = 4GM \ln\left(\frac{1}{u}\right) + C + O(u).$$
A distant observer probes the radiation emerging from the black hole by measuring a quantum field Ψ. Assume they measure this as a function of t and Ω at some fixed distance. A typical observable is a two-point function
$$\langle \Psi(\Omega, t)\Psi(\Omega', t')\rangle.$$
In a spherically symmetric Schwarzschild background (indeed, even Kerr, if you exploit the fact that the wave equations for Kerr are separable), Ψ admits a partial wave expansion.



- To get at the essence of the phenomenon, consider a particular partial wave ψ of Ψ, and assume it is (e.g.) a chiral free fermion. In the vacuum its two-point function is

$$\langle \psi(u)\psi(u')\rangle = \frac{(du\, du')^{1/2}}{u-u'},$$

  and then by the expression above (discarding the $O(u)$ terms) we get

$$\langle \psi(t)\psi(t')\rangle = \frac{1}{4GM}\frac{(dt\, dt')^{1/2}}{e^{(t-t')/8GM} - e^{(t-t')/8GM}}.$$

  This is antiperiodic in imaginary time; it is odd under $t \to t + 8\pi GMi$. This corresponds to a thermal correlation function with temperature

$$T_H = \frac{1}{8\pi GM}.$$

  We call this the *Hawking temperature* of the black hole; the BH radiates thermally at this temperature.

1.2 The Smarr relation for nonzero cosmological constant

- Notes from Kastor et al. [2]. This is a very technical paper – I've spelled out many more details than I understand, but because the results here allow us to define a thermodynamic volume and pressure for black holes.

- We would like to derive the Smarr relation for cases where $\Lambda \neq 0$. One typically invokes Euler's theorem for homogeneous functions,

$$f(\alpha^p x, \alpha^q y) = \alpha^r f(x,y)$$

  since this implies by the chain rule that

$$rf(x,y) = p\left(\frac{\partial f}{\partial x}\right)x + q\left(\frac{\partial f}{\partial y}\right)y.$$

- In the context of black hole thermodynamics, we work with a black hole mass M that depends on A and Λ, and we would like to argue for homogeneity. This is done using a dimensional argument. In D spacetime dimensions, one has

$$S = \frac{1}{8\pi G}\int d^D x\, \sqrt{-g}(R - 2\Lambda),$$

which implies that $[\Lambda] = (\text{length})^{-2}$. One also argues that $[M] = (\text{length})^{D-3}$ and $[A] = (\text{length})^{D-2}$. Assuming that this fixes homogeneity, we get

$$(D-3)M = (D-2)\left(\frac{\partial M}{\partial A}\right)A - 2\left(\frac{\partial M}{\partial \Lambda}\right)\Lambda.$$

This is just a scaling argument, but it is further supported by the fact that you can get the same result for $\Lambda = 0$ using something called a Komar integral relation, assuming you have a static/stationary black hole in vacuum Einstein gravity.

- The Komar integral method does not go through for $\Lambda \neq 0$, for then the field equations imply

$$R_{ab} = \frac{2\Lambda}{D-2}\, g_{ab},$$

and in particular, a Killing vector satisfies

$$\nabla_a \nabla^a \xi^b = -\frac{2\Lambda}{D-2}\,\xi^b,$$

so there is a nonvanishing boundary term which spoils the argument. However, Bazanski and Zyla [3] and Kastor [4] show that one can construct a new Komar integral relation by addition of a new term to make this vanish:

$$\frac{D-2}{8\pi G}\int_{\partial\Sigma} dS_{ab}\left(\nabla^a \xi^b + \underbrace{\frac{2}{D-2}\Lambda\omega^{ab}}_{\text{new term}}\right) = 0,$$

where ω^{ab} is a Killing potential associated with ξ^b via $\xi^b = \nabla_a \omega^{ab}$. This potential is not unique and can be changed by adding a term $\omega'^{ab} = \omega^{ab} + \lambda^{ab}$ so long as $\nabla_a \lambda^{ab} = 0$. But it is shown that the result doesn't depend on λ^{ab}. (This isn't quite correct for a technical reason; there is something said about how only the difference of integrals of ω^{ab} over the inner and outer boundaries is physically meaningful, similar to how only differences in electric potential are.)

1.3 The first law for varying cosmological constant; mass as enthalpy

- Notes from Kastor et al. [2], which uses a method by Sudarsky and Wald [5] to derive a version of the first law that includes variations in Λ. Very technical; the idea is to apply Hamiltonian perturbation theory to a static, asymptotically AdS black hole with bifurcate Killing horizon. One obtains

$$\delta M = \frac{\kappa}{8\pi G}\,\delta A + \frac{\Theta}{8\pi G}\,\delta\Lambda,$$

where the quantity

$$\Theta = -\left[\int_{\partial\Sigma_\infty} dS_{ab}(\omega^{ab} - \omega^{ab}_{\text{AdS}}) - \int_{\partial\Sigma_h} dS_{ab}\,\omega^{ab}\right]$$

is (minus) the difference between the integral of the renormalized Killing potential at infinity and the integral of the Killing potential on the horizon. This quantity is given the interpretation

$$\Theta = V_{\text{BH}} - V_{\text{AdS}},$$

the difference of two (infinite) volumes; the idea is that $V = -\Theta$ gives a measure of the volume excluded from the spacetime by the black hole horizon. If we then think of the cosmological constant as providing a thermodynamic pressure $P = -\Lambda/8\pi G$, we may interpret the first law as the statement that

$$\delta M = T\delta S + V\delta P.$$

This coincides with the variation of the enthalpy H in classical thermodynamics; thus, it is argued that the mass of an AdS black hole should be thought of as the enthalpy $H = E + PV$ of the spacetime.

- A physical argument to support this: if you naively integrate the energy density between the black hole horizon and infinity, then you get an infinite result, because the cosmological constant carries an energy density of $\rho = \Lambda/8\pi G$. Adding a (positive) PV term cancels out this negative ρV contribution to the energy and yields a finite result.

1.4 Reversible and irreversible transformations

- Notes from Christodoulou [6] and Christodoulou and Ruffini [7]. The first paper introduces the concepts of irreducible mass M_{irr} and reversible/irreversible transformations for Kerr black holes. The latter presents a formula for the irreducible mass when the black hole charge is nonzero. These results are used to derive efficiency bounds on energy extraction processes. **(reading list)**

- It's important to note that the irreducible mass and horizon area are related; the statements $dM_{\text{irr}} \geq 0$ and $dA \geq 0$ are equivalent.

2 Thermodynamic geometry

- Salamon et al. [8] describe the group of coordinate transformations which preserve both the first law of thermodynamics and the Weinhold metric, listing out the generators and corresponding commutation relations. The discussion is phrased in group-theoretic language; for instance, the generator τ corresponds to a classical Legendre transformation.

2.1 Proof that the Ruppeiner and Weinhold metrics are conformal

- Notes from Salamon et al. [9] and Callen [10]. The Ruppeiner and Weinhold metrics describe different geometries, but it is very useful, for computational purposes, to note that they are conformal:
$$g^R = \frac{1}{T}\, g^W.$$
To prove this, we need the definition of temperature,
$$\frac{\partial S}{\partial U} = \frac{1}{T},$$
as well as the first law of thermodynamics,
$$dU = T\, dS - p\, dV + \mu\, dN.$$
This can be generalized by inserting additional work terms; the same argument will go through. First note that the Ruppeiner metric can be written
$$\begin{aligned} g^R &= -\frac{\partial^2 S}{\partial X^\alpha \partial X^\beta}\, dX^\alpha\, dX^\beta \\ &= -dX^\mu\, dY_\mu, \end{aligned}$$
where $X^\mu = (U, V, N)$ are the natural variables of S, and we've defined $Y_\mu := \partial S/\partial x^\mu$. By rewriting the first law in the form
$$dS = \frac{1}{T}\, dU + \frac{p}{T}\, dV - \frac{\mu}{T}\, dN,$$
it is straightforward to see that $Y_\mu = (1/T,\, p/T, -\mu/T)$. Substituting back, we get
$$\begin{aligned} g^R &= -d\left(\frac{1}{T}\right) dU - d\left(\frac{p}{T}\right) dV + d\left(\frac{\mu}{T}\right) dN \\ &= \frac{1}{T^2}\, dT\, dU - \frac{1}{T}\, dP\, dV + \frac{p}{T^2}\, dT\, dV + \frac{1}{T}\, d\mu\, dN - \frac{\mu}{T^2}\, dT\, dN \\ &= \frac{1}{T^2}\, dT\big(T\, dS - \cancel{p\, dV} + \cancel{\mu\, dN}\big) - \frac{1}{T}\, dP\, dV + \cancel{\frac{p}{T^2}\, dT\, dV} + \frac{1}{T}\, d\mu\, dN - \cancel{\frac{\mu}{T^2}\, dT\, dN} \\ &= \frac{1}{T}(dT\, dS - dP\, dV + d\mu\, dN). \end{aligned}$$
But similarly to the Ruppeiner metric, the Weinhold metric can be written
$$\begin{aligned} g^W &= -dX^\mu\, dY_\mu \\ &= dT\, dS - dP\, dV + d\mu\, dN, \end{aligned}$$

where $X^\mu = (S, V, N)$ are the natural variables of U and $Y_\mu = (T, -P, \mu)$ are the conjugates in the energy representation. This shows that

$$g^R = \frac{1}{T} g^W,$$

as we wanted. Nothing was assumed about extensivity or concavity/convexity of the potentials, so the same relation holds in black hole thermodynamics. In particular, either the black hole mass or entropy can be used to compute both metrics.

2.2 Extensivity

2.2.1 Renyi and Tsallis entropy

- Notes from Promsiri et al. [11]. It is noted that the black hole entropy, as derived via the Gibbs-Boltzmann statistical approach (these authors' term), appears to be nonextensive. For example, when two black holes of entropies S_A and S_B merge adiabatically, the resulting entropy of the combined system S_AB does not follow an additive law. One proposal, by Abe (2001), is that this entropy follows a generalized nonadditive composition rule

  $$H(S_\mathrm{AB}) = H(S_\mathrm{A}) + H(S_\mathrm{B}) + \lambda H(S_\mathrm{A}) H(S_\mathrm{B}),$$

  where $H(S)$ is a differentiable function of S and λ is a nonextensivity parameter. Then the usual statistical mechanics arguments allow for the determination of macroscopic (i.e. thermodynamic) properties.

- One reason that the GB entropy formula might not be appropriate for gravitating systems is that long-range interactions cannot be ignored; this may explain where nonextensivity comes from.

- It turns out that another entropic functional form,

  $$L(S) = \frac{1}{\lambda} \ln[1 + \lambda H(S)],$$

  makes the composition rule additive,

  $$L(S_\mathrm{AB}) = L(S_\mathrm{A}) + L(S_\mathrm{B}).$$

  If one identifies the differentiable function $H(S)$ to be the Tsallis entropy $H(S) = S_\mathrm{T}$, then one obtains the Renyi entropy,

  $$L(S_\mathrm{T}) = \frac{1}{\lambda} \ln(1 + \lambda S_\mathrm{T}) \equiv S_\mathrm{R}.$$

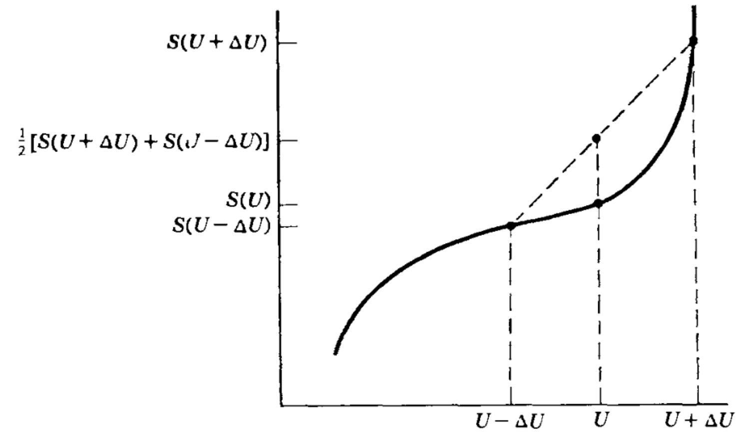
  This supposedly allows for a well-defined notion of a thermal equilibrium state for nonextensive systems, and uniquely determines a temperature function from the usual relation

  $$\frac{1}{T_\mathrm{R}} = \frac{\partial S_\mathrm{R}}{\partial E}.$$

- It may not be worth reading this paper; the authors claim that the Ricci scalar for the Reissner-Nordstrom black hole is nonzero, which is false. They cite Shen et al. (2007) for this claim; I know that there is at least one paper which presents an incorrect expression for the Ruppeiner metric. It may be this one!

2.3 Thermodynamic stability

- Notes from Chapter 8 of Callen [10].
- Suppose we have two identical subsystems, each with fundamental equation $S = S(U, V, N)$. The two are initially separated by an impermeable partition, but then are allowed to exchange energy. Suppose further that S has the form below.



  According to the entropy maximum principle, the system is not stable at this point. To see why, suppose that an amount of energy ΔU is exchanged between the two subsystems. Then the entropy of the new configuration is larger than the original:

$$S(U + \Delta U) + S(U - \Delta U) > 2S(U, V, N).$$

  To maximize entropy, the subsystems would spontaneously exchange energy. If this entropy function represented a single, undivided system, this argument shows that inhomogeneities would develop within the system.

- In other words, the condition of stability is the following:

$$S(U + \Delta U) + S(U - \Delta U) \leq 2S(U, V, N) \qquad \text{(for all } \Delta U\text{)}$$

  The corresponding differential statement (which is weaker) is

$$\left(\frac{\partial^2 S}{\partial U^2}\right)_{V,N} \leq 0.$$

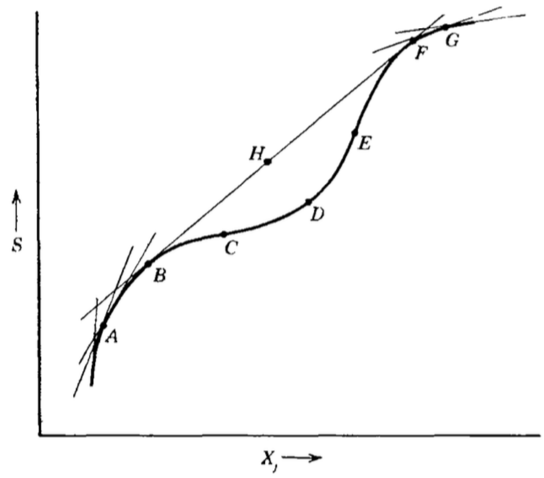
  The same arguments apply for the extensive variables V and N.

- However, one might obtain a fundamental equation that does not satisfy these conditions, either via a statistical mechanical calculation, or by extrapolating from experimental data. In these cases, one can obtain the stable fundamental equation by drawing an envelope of tangent lines that lie above the original curve:

  In this example, the portion $BCDEF$ is replaced by the line BHF. Notice that both the differential and global conditions fail at point D, whereas points B and E satisfy the differential condition, but fail the global one. One says that the portions BC and EF are locally stable, but globally unstable. (Other texts use *metastable*.) Points on the straight portion correspond to a phase separation, with part of the system in state B, and part in state F.

- These considerations apply when only one variable X_j is allowed to vary; if both U and V can change, then the entropy surface must lie everywhere below its tangent planes. One requires

$$S(U + \Delta U, V + \Delta V, N) + S(U - \Delta U, V - \Delta V, N) \leq 2S(U, V, N),$$



for arbitrary ΔU and ΔV, which implies the previous condition, as well as

$$\frac{\partial^2 S}{\partial U^2}\frac{\partial^2 S}{\partial V^2} - \left(\frac{\partial^2 S}{\partial U \partial V}\right)^2 \geq 0.$$

This last condition rules out the possibility of a "fluted" entropy surface which curves downward along the four directions $\pm U$ and $\pm V$, but curves up along the diagonal directions between the U and V axes.

- Since thermodynamic response functions (e.g. heat capacities) are expressible as derivatives of the fundamental equations, the stability conditions impose restrictions on these functions. For example, since

$$\left(\frac{\partial^2 S}{\partial U^2}\right)_{V,N} = -\frac{1}{NT^2 c_v} \leq 0,$$

it follows that the molar heat capacity must be positive in a stable system. Similar arguments using the convexity of the Helmholtz potential with respect to the volume show that $\kappa_T > 0$. Using the identities

$$c_p - c_v = \frac{Tv\alpha^2}{\kappa_T}$$

and

$$\frac{\kappa_s}{\kappa_T} = \frac{c_v}{c_p},$$

it follows that stability requires

$$c_p \geq c_v \geq 0, \qquad \kappa_T \geq \kappa_s \geq 0.$$

In other words, the addition of heat at constant pressure or at constant volume necessarily increases the temperature of a stable system. Decreasing the volume, either isothermally or isentropically, necessarily increases the pressure of a stable system. Neither of these are plausibility arguments – they follow from the entropy maximum principle.

- In general, in an $r+2$ dimensional thermodynamic space $(S, X_0, X_1, \ldots, X_r)$, stability requires that the entropy hypersurface lie everywhere below its family of tangent hyperplanes.

- These conditions imply that the Ruppeiner metric is positive-definite in single-phase regions. In general, we expect one or more of these conditions to be violated at a phase transition. The same statement applies to the Weinhold metric.

2.4 Phase transitions

2.5 Thermodynamic length

2.6 Geodesics as dissipation and entropy-minimizing curves

- Notes from Salamon and Berry [12].

2.7 Ruppeiner curvature and interactions

- Ruppeiner [13] argues that the scalar curvature R is a measure of interactions. The idea is to plot R over your state space, then regions with negative (positive) R have attractive (repulsive) interactions. This is very easy to test in practice, but there is little theoretical basis for this interpretation. We also know of a few systems which do not fit into this classification. García-Ariza et al. [14] showed that the ideal gas is not uniquely characterized by $R = 0$, by constructing non-ideal gas response functions which lead to vanishing curvature.

- There is an ambiguity in the definition of the Ruppeiner metric that goes underappreciated in the literature. If one were to allow all natural variables (U, V, N) of the entropy to fluctuate, then the Gibbs-Duhem relation would imply degeneracy of the metric. To avoid this, one can hold an extensive variable constant. The choice of variable to hold constant amounts to selecting a different hypersurface on which to induce the metric; different choices yield different geometries. Some studies will pick one or the other without specifying their choice!

- The fixed variable is typically the volume V (it has been argued that this choice is attractive for theoretical reasons), but De Leon and Vega [15] showed that the particle number N appears to be useful for the study of phase transitions. In particular, the geodesics of the N-metric appear to be sensitive to the partitioning of the state space into distinct phases, even in the supercritical region.

- We [16] continue this theme of comparing the two induced geometries, and find that the ideal gas appears to be privileged; we argue that it is the only system for which both V and N metrics are flat. On the basis of this finding, we argue that one should take both induced metrics into account when probing the nature of interactions in a system.

- It seems as though you should be able to define a U-metric, as well. **(why not?)**

- This issue of selecting a hypersurface only arises due to the Gibbs-Duhem relation, which in turn follows from the homogeneity of the entropy,

$$S(\lambda U, \lambda V, \lambda N) = \lambda S(U, V, N).$$

  This always holds in ordinary thermodynamics, but sometimes you have an entropy function that is not homogeneous, e.g. in black hole thermodynamics.

3 Kerr-Newman

- The discussion here follows Ruppeiner [17, 18].
- The Kerr-Newman black hole has entropy given by

$$S(M,J,Q) = \frac{1}{8}\Big(2M^2 - Q^2 + 2\sqrt{M^4 - J^2 - M^2Q^2}\Big);$$

compare Davies [19] and Smarr [20]. In principle, one can examine (M,J,Q), (J,Q), (M,Q), and (M,J) fluctuations. The case of only one fluctuating variable yields a flat geometry; the curvature scalar always vanishes in one dimension.

- The physical regime for the Kerr-Newman black hole is where

$$\left(\frac{J}{M^2}\right)^2 + \left(\frac{Q}{M}\right)^2 < 1;$$

the extremal limit corresponds to letting the left-hand side approach unity. If we introduce the dimensionless variables

$$\alpha = \frac{J^2}{M^4}, \qquad \beta = \frac{Q^2}{M^2},$$

then the same inequality can be written

$$\alpha + \beta < 1.$$

These variables were used in the analysis of Davies [19]. One can also introduce

$$K = \sqrt{1-\alpha-\beta}, \qquad L = \sqrt{1+\alpha},$$

in terms of which the Hawking temperature T can be written

$$\frac{1}{T} = \frac{(K^2 + 2K + L^2)M}{4K}.$$

One can see that the extremal limit $K = 0$ corresponds to zero temperature, which is thought to be unattainable by the third law of black hole mechanics.

3.1 Kerr

- Notes from Aman et al. [21]. The coordinate transformation described here is also used in [22].
- Using the Smarr formula for the outer horizon of the black hole,

$$S = 2M^2\left(1 + \sqrt{1 - \frac{J^2}{M^4}}\right),$$

we compute the Ruppeiner metric, and obtain

$$g^R = -\frac{4}{(1-J^2/M^4)^{3/2}}\left[1 - \frac{3J^2}{M^4} + \left(1 - \frac{J^2}{M^4}\right)^{3/2}\right] dM^2 - \frac{8J}{M^3(1-J^2/M^4)^{3/2}}\, dM\, dJ$$
$$+ \frac{2}{M^2(1-J^2/M^4)^{3/2}}\, dJ^2,$$

Notice that the metric becomes singular in the extremal limit $J \to M^2$. The physical state space is bounded by the parabolas $J = \pm M^2$, which mark extremal black holes.

- This expression isn't very enlightening, so we diagonalize the metric by defining the coordinate

$$a = \frac{J}{M^2}.$$

This is constrained to $-1 \le a \le 1$; the extremal parabolas get mapped to straight lines. Then

$$g^R = -4\left(1 + \frac{1}{\sqrt{1-a^2}}\right) dM^2 + \frac{2M^2}{(1-a^2)^{3/2}}\, da^2.$$

Since in these coordinates

$$T = \frac{1}{4M}\frac{\sqrt{1-a^2}}{1+\sqrt{1-a^2}},$$

the Ruppeiner metric can also be written

$$g^R = \frac{1}{T}\left[-\frac{dM}{M^2} + \frac{M}{2}\frac{da^2}{(1-a^2)\left(1+\sqrt{1-a^2}\right)}\right].$$

Then the Weinhold metric is the expression in square brackets:

$$g^W = -\frac{dM^2}{M} + \frac{M}{2}\frac{da^2}{(1-a^2)\left(1+\sqrt{1-a^2}\right)}.$$

This metric is flat, as can be verified by computing the Ricci scalar and showing that it is zero. (This only works for two-dimensional metrics; in general, you need to show that the Riemann tensor is zero.)

- We then perform a series of coordinate transformations to bring the metric to manifestly flat form. Since $-1 \le a \le 1$, we can introduce

$$a = \sin 2\beta,$$

with the constraint $-\pi/4 \le \beta \le \pi/4$. This puts the metric in the form

$$g^W = -\frac{dM^2}{M} + \frac{M}{\cos^2\beta}\, d\beta^2.$$

Then, defining $\tau = 2M^{1/2}$, we obtain

$$g^W = -d\tau^2 + \frac{\tau^2}{4\cos^2\beta}\, d\beta^2.$$

(finish)

3.2 Reissner-Nordström

- Reissner-Nordström black holes have a flat Ruppeiner metric. A quick way to see this is to set $J = 0$ in the expression for the Kerr-Newman entropy and then calculate the curvature scalar for (M, Q) fluctuations. The result is that $R = 0$, which implies a flat metric. This result was first noted by Aman et al. [21] and is discussed further in Mirza and Zamani-Nasab [23]. Please check, for consistency, that the same result is obtained whether you first compute the metric in (M, J, Q) space and then set $J = 0$, or you start by setting $J = 0$, and then compute the metric in the restricted (M, Q) space.

- Alternatively, one can show explicitly that the metric is flat. Aman et al. [21] obtained an expression for the Ruppeiner metric of the RN black hole. By using the fundamental relation (an inverted form of the Smarr relation)

$$M = \frac{\sqrt{S}}{2}\left(1 + \frac{Q^2}{S}\right)$$

to first calculate the Weinhold metric, they obtained

$$g_W = \frac{1}{8S^{3/2}}\left[-\left(1 - \frac{3Q^2}{S}\right)dS^2 - 8Q\,dQ\,dS + 8S\,dQ^2\right].$$

They then introduce the coordinate $u \equiv Q/\sqrt{S}$, which is restricted to the physical (i.e. non-extremal) range $-1 \leq u \leq 1$. This diagonalizes the metric:

$$g_W = \frac{1}{8S^{3/2}}\left[-(1-u^2)\,dS^2 + 8S^2\,du^2\right].$$

Using the conformal relationship between the Weinhold and Ruppeiner metrics, they get

$$g_R = -\frac{dS^2}{2S} + 4S\,\frac{du^2}{1-u^2}.$$

Finally, changing coordinates to (τ, σ) via $\tau = \sqrt{2S}$ and $\sin(\sigma/\sqrt{2}) = u$ yields the metric

$$g_R^2 = -d\tau^2 + \tau^2\,d\sigma^2,$$

which is manifestly flat.

- Mirza and Zamani-Nasab [23] argue that the proper way to study the Ruppeiner geometry of RN black holes is to start from the complete set of thermodynamic variables, i.e., one should first compute the curvature scalar for the Kerr-Newman solution in AdS, and then take the $J \to 0$, $l \to \infty$ limit. This yields the result

$$R = \frac{S^2 + Q^2 S + 2Q^4}{(S^2 - Q^4)(S + Q^2)},$$

which is notable because the curvature scalar is nonzero and diverges at the extremal limits $Q = \pm\sqrt{S}$.

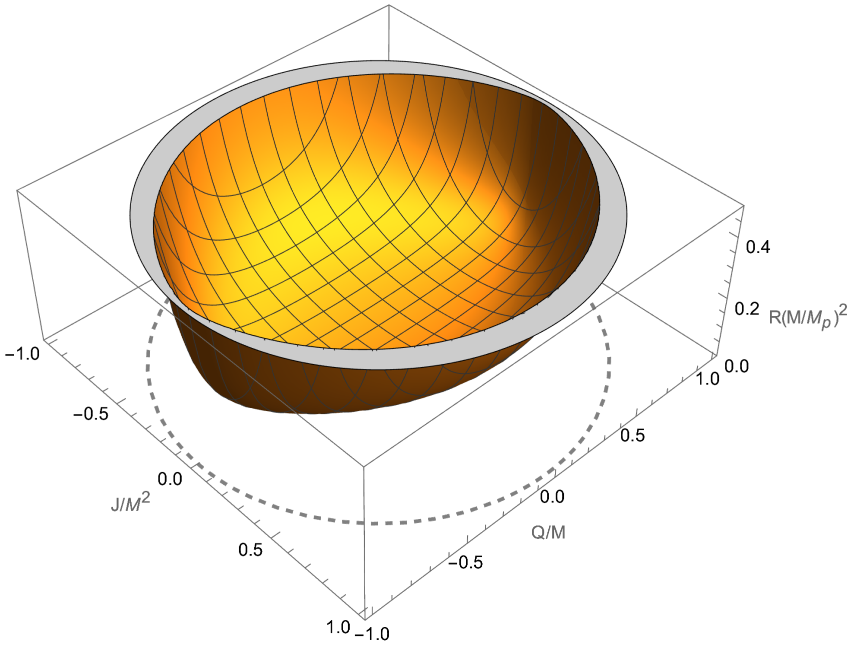


Figure 1: The rescaled curvature scalar $R(M/M_p)^2$ of the Kerr-Newman black hole, plotted for (M, J, Q) fluctuations. The extremal limit is shown as a gray dashed curve. The curvature scalar is strictly positive in the physical regime, attains a minimum value of $M_p^2/4\pi M^2$ at the origin, and diverges as T^{-1} in the extremal limit.

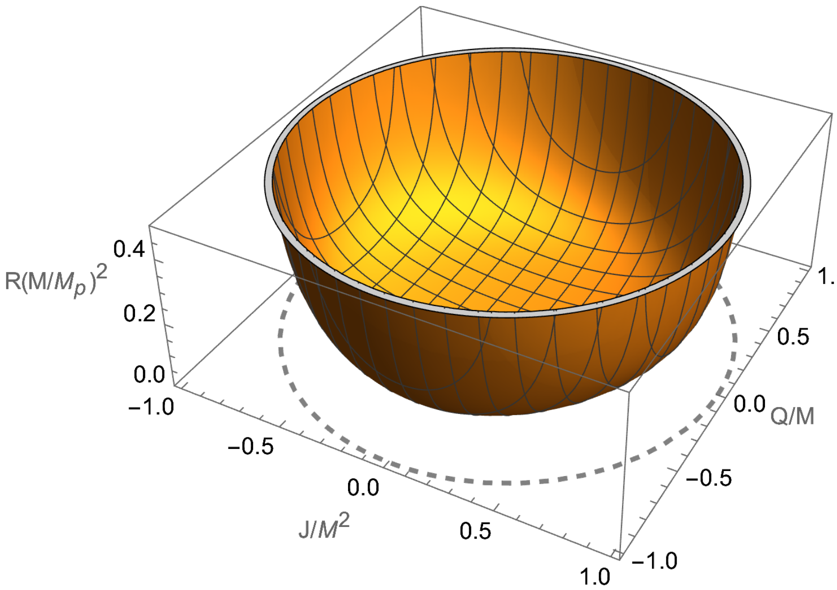


Figure 2: The rescaled curvature scalar $R(M/M_p)^2$ of the Kerr-Newman black hole for (J, Q) fluctuations. The curvature scalar is positive everywhere except at the origin, where it is zero, and diverges as T^{-1} in the extremal limit.

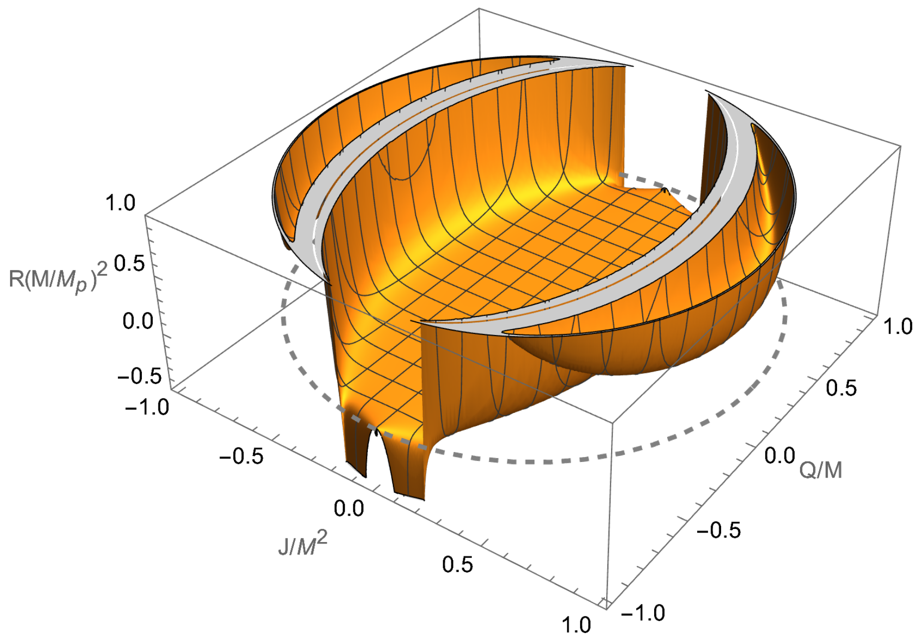


Figure 3: The rescaled curvature scalar $R(M/M_p)^2$ of the Kerr-Newman black hole for (M,Q) fluctuations. There are two regimes of negative R near the extremal values $Q=\pm M$. The curvature scalar is zero along the line $J=0$. This is consistent with the observation of Aman et al. [21] that Reissner-Nordström black holes have zero curvature.

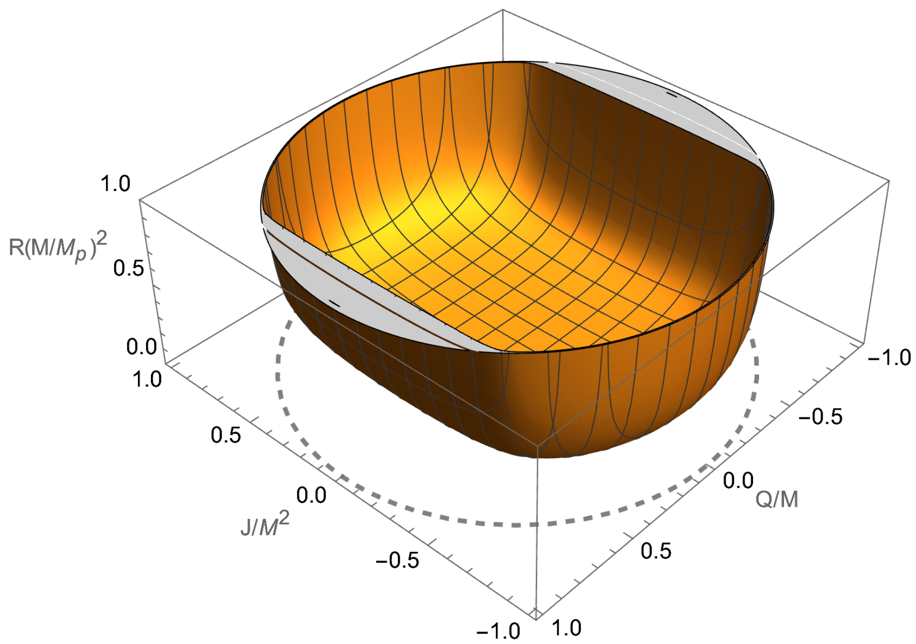


Figure 4: The rescaled curvature scalar $R(M/M_p)^2$ of the Kerr-Newman black hole for (M,J) fluctuations. This is positive everywhere in the physical regime, and diverges both in the extremal limit and at the limits of stability.

4 RN-AdS

- The discussion of the thermodynamic geometry follows Wei et al. [24, 25]. For future reference, there are also a couple of interesting papers on the thermodynamics of RN-AdS by Louko and Winters-Hilt [26] and Chamblin et al. [27].

- The inclusion of a nonzero cosmological constant allows us to identify a thermodynamic volume for the system (see Kubiznak and Mann [28]; also the review article by Kubiznak et al. [29]). In general, in d spacetime dimensions, one can identify a pressure

$$P = -\frac{1}{8\pi}\Lambda = \frac{(d-1)(d-2)}{16\pi}\frac{1}{\ell^2},$$

with conjugate variable

$$V = \left(\frac{\partial M}{\partial P}\right)_{S,\,Q_i,\,J_k}$$

We will specialize to the case $d = 4$, so that

$$P = \frac{3}{8\pi}\frac{1}{\ell^2}.$$

For a RN-AdS black hole, we have

$$V = \frac{4}{3}\pi r_+^3,$$

where r_+ is the radius of the black hole event horizon in terms of the Schwarzschild radial coordinate. By the "Euclidean trick,"[1] the black hole temperature is

$$T = \frac{1}{4\pi r_+} + 2r_+P - \frac{Q^2}{4\pi r_+^3},$$

with corresponding entropy

$$S = \pi r_+^2.$$

(why no P dependence?) Note, for later, that the pressure equation of state can be written

$$P = \frac{T}{2r_+} - \frac{1}{8\pi r_+^2} + \frac{Q^2}{8\pi r_+^4}.$$

- We can compute the heat capacities at constant volume and pressure.

$$\begin{aligned}
C_P &= T\left(\frac{\partial S}{\partial T}\right)_{P,Q} \\
&= \left(\frac{1}{4\pi r_+} + 2r_+P - \frac{Q^2}{4\pi r_+^3}\right)(2\pi r_+)\left(-\frac{1}{4\pi r_+^2} + 2P + \frac{3Q^2}{4\pi r_+^4}\right)^{-1} \\
&= 2S\frac{8PS^2 + S - \pi Q^2}{8PS^2 - S + 3\pi Q^2}.
\end{aligned}$$

$$\begin{aligned}
C_V &= T\left(\frac{\partial S}{\partial T}\right)_{V,Q} \\
&= 0.
\end{aligned}$$

[1] No sources were cited here, but a search suggests that this method is outlined in Hawking and Page [30]. This paper is an early study of chargeless, spinless black holes with $\Lambda < 0$.

One can see that $C_V = 0$ since S depends on V alone. This is problematic, because it can be shown (see, e.g., Section 2.1 of undergrad thesis) that the Ruppeiner metric for fixed Q is

$$g_Q = \frac{C_V}{T^2}\,dT^2 - \frac{1}{T}\left(\frac{\partial P}{\partial V}\right)_T dV^2.$$

Thus, the metric is singular. Wei et al. [24] propose setting C_V to some constant value for the purposes of the calculation, at the end taking the limit

$$C_V \to 0^+.$$

The curvature scalar diverges in this limit, so they examine the normalized curvature scalar

$$\tilde{R} = RC_V.$$

This procedure is motivated by noting that $C_V = \frac{3}{2}k_B \sim 10^{-23}$ J/K for the van der Waals gas, so this process resembles taking a $k_B \to 0^+$ limit.[2] The problem persists if we instead fix the thermodynamic volume V to examine the metric g_V, because the metric component g_{TT} is still proportional to C_V:

$$g_V = \frac{C_V}{T^2}\,dT^2 + \frac{1}{T}\left(\frac{\partial \Phi}{\partial Q}\right)_{T,V} dQ^2.$$

By the way, recall that RN was also unusual in asymptotically flat spacetime; in that case, the thermodynamic metric is flat.

- We note that for this system, g_V and g_Q are the black-hole analogues of the constant volume and constant particle number metrics in ordinary thermodynamics.

4.1 Thermodynamic curvature

- From the pressure EoS, we can identify the critical values

$$P_c = \frac{1}{96\pi Q^2}, \qquad T_c = \frac{\sqrt{6}}{18\pi Q}, \qquad V_c = 8\sqrt{6}\pi Q^3, \qquad v_c = 2\sqrt{6}Q,$$

which we will use for defining dimensionless variables. Since we are assuming a constant heat capacity, the scalar curvature takes the form [this is Eq. (48) in Wei]

$$R = \frac{(\partial_x y)^2 - T^2(\partial_{T,x}y)^2 + 2T^2(\partial_x y)(\partial_{T,T,x}y)}{2C_x(\partial_x y)^2},$$

where x is the extensive variable that is allowed to fluctuate, and y is the corresponding intensive variable. If we use the constant-Q metric and define dimensionless variables

$$\tilde{P} = P/P_c, \qquad \tilde{V} = V/V_c, \qquad \tilde{T} = T/T_c,$$

and so on, then we obtain the normalized curvature scalar

$$\tilde{R}_Q = -\frac{(-1 + 3\tilde{V}^{2/3})(1 - 3\tilde{V}^{2/3} + 4\tilde{T}\tilde{V})}{2(1 - 3\tilde{V}^{2/3} + 2\tilde{T}\tilde{V})^2}.$$

As a check on this result, the same expression is obtained using either (48), as given above, or by directly computing the curvature scalar from the metric components. A plot of the curvature scalar is given below.

[2] Some definitions of the Ruppeiner metric have a factor of $1/k_B$ that cancels out the k_B from C_V. Then one has $g_{TT} = 3/2T^2$, which does not vanish under the limiting procedure. What does a $k_B \to 0^+$ limit correspond to? Is it physically motivated, or purely formal?

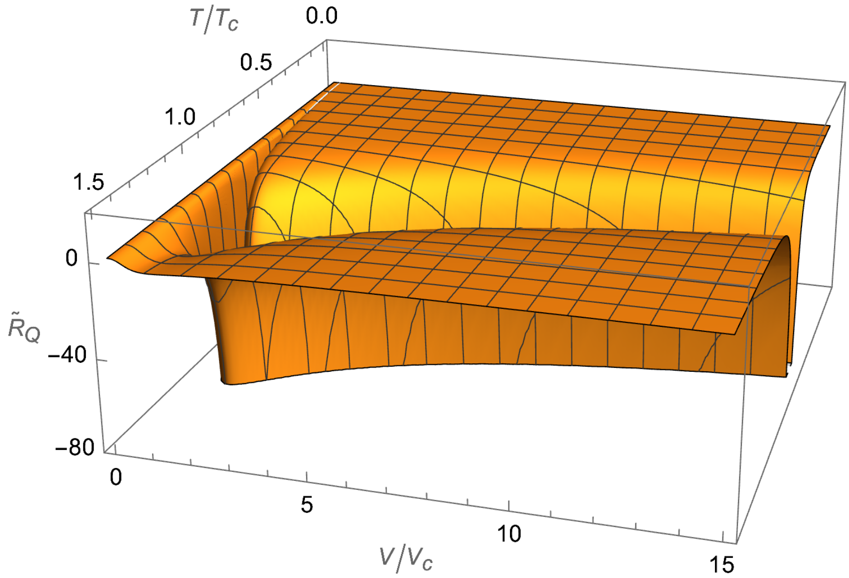


Figure 5: The normalized curvature scalar $\tilde{R}_Q = R_Q C_V$ for the RN-AdS black hole with fixed charge Q. This is negative over most of the parameter space, but there are two sign-changing curves. The curvature scalar diverges along the curve $\tilde{T} = (3\tilde{V}^{2/3} - 1)/2\tilde{V}$, which includes the critical point.

4.2 Black hole-fluid analogy

The black hole-fluid analogy boils down to the observation that the black hole pressure equations of state sometimes resemble those of fluids. This requires nonzero cosmological constant, because varying Λ is what allows us to identify a thermodynamic volume for the black hole.

4.2.1 The vdW-like character of RN-AdS

- Notes from Kubiznak and Mann [28].
- Rewrite the expression for the black hole temperature of RN-AdS in the form

$$P = \frac{T}{2r_+} - \frac{1}{8\pi r_+^2} + \frac{Q^2}{8\pi r_+^4}.$$

We would like to make an explicit identification with the van der Waals equation of state,

$$Pv^3 - (kT + bP)v^2 + av - ab = 0.$$

Multiplying our black hole pressure P and temperature T by appropriate conversion factors to give them the dimensions of physical pressure and temperature, respectively, we have

$$\text{Press} = \frac{\hbar c}{L_p^2} P, \qquad \text{Temp} = \frac{\hbar c}{k} T,$$

where L_p is the Planck length. We then get

$$\frac{\hbar c}{L_p^2} P = \frac{k}{2L_p^2 r_+}\left(\frac{\hbar c}{k} T\right) + \cdots$$

which, upon comparison with the van der Waals equation, motivates the identification

$$v = 2L_p^2 r_+.$$

Thus, in this black hole-fluid analogy, it is the horizon radius r_+ and not the thermodynamic volume $V = 4\pi r_+^3/3$ which should be identified with the specific volume v. However, when computing the Ruppeiner metric, we return to the thermodynamic volume V. The role of v here is to allow us to identify a critical point for the RN-AdS black hole.

- Although not explicitly stated, their argument seems to be that the pressure equation of state should be expressible as a polynomial in v, and a naive identification of V with v would give us a pressure EoS of the form $Pv^{4/3} + \cdots = 0$. It is not hard to show that this EoS exhibits no critical behavior.

- The previous argument does not fix the numerical prefactor of r_+, but it is shown that this assignment leads to the result
$$\frac{P_c v_c}{T_c} = \frac{3}{8},$$
which is notable because the same dimensionless ratio is obtained for van der Waals.

- With this, the pressure equation of state can be written (in geometric units)
$$P = \frac{T}{v} - \frac{1}{4\pi v^2} + \frac{2Q^2}{\pi v^4}.$$
Unlike the van der Waals equation of state, this pressure equation of state is quartic in v, but the isotherms (for fixed Q) look similar, and are reproduced here.

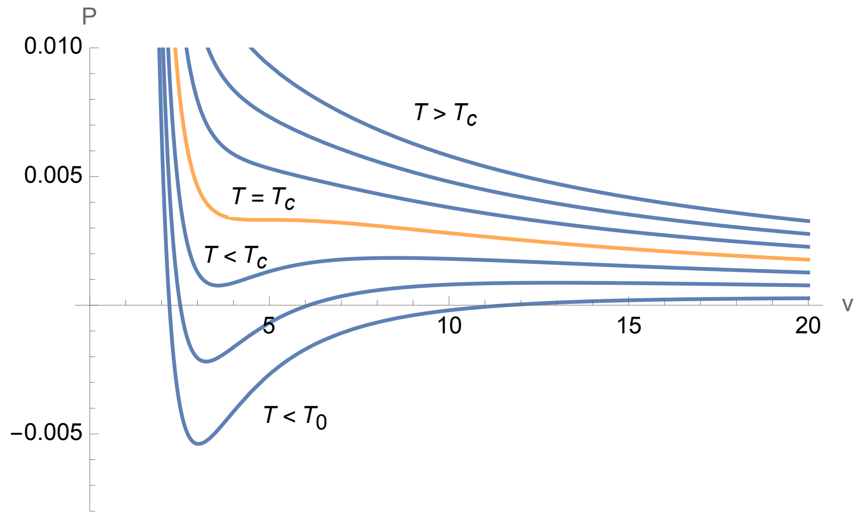


Figure 6: Isotherms for the charged AdS solution with $Q = 1$. There is a critical temperature T_c above which inflection points no longer exist, as well as a temperature T_0 below which the isotherms exhibit unphysical negative pressures. Similar phenomena appear in the isotherms of the van der Waals gas.

- The critical point, where both the first- and second-order derivatives of P with respect to v are zero, is located at
$$T_c = \frac{\sqrt{6}}{18\pi Q}, \qquad v_c = 2\sqrt{6}Q, \qquad P_c = \frac{1}{96\pi Q^2}.$$
The temperature T_0 below which the pressure is negative for certain ranges of v is given by
$$T_0 = \frac{\sqrt{3}}{18\pi Q}.$$

Interestingly, the existence of both a critical point and the temperature T_0 requires a spherical topology for the black hole horizon, for in general one obtains

$$P = \frac{T}{v} - \frac{k}{2\pi v^2} + \frac{2Q^2}{\pi v^4},$$

where $k = 0$ is the toroidal (planar) case and $k = -1$ is the higher-genus (hyperbolic) case. In either of these cases, all terms on the right-hand side are positive, and P is positive everywhere and monotonically decreasing.

4.2.2 Van der Waals black hole

- Rajagopal et al. [31] push the black hole-fluid analogy further by designing an asymptotically AdS metric whose pressure equation of state is formally identical to van der Waals. They show that their solution satisfies all three energy conditions at low pressures, and that the dominant and weak energy conditions are violated at large pressures (the strong condition is always satisfied). They start with the static spherically symmetric ansatz

$$ds^2 = -f\,dt^2 + \frac{dr^2}{f} + r^2\,d\Omega^2$$

with

$$f = \frac{r^2}{\ell^2} - \frac{2M}{r} - h(r, P),$$

where ℓ is related to the cosmological constant by $\Lambda = -3/\ell^2$, and h is a function to be determined. The corresponding energy density and principal pressures are

$$\rho = -p_1 = \frac{1 - f - rf'}{8\pi r^2} + P,$$

$$p_2 = p_3 = \frac{rf'' + 2f'}{16\pi r} - P.$$

The approach is to first choose f so that the pressure EOS is van der Waals, and then check whether this solution satisfies the energy conditions.

- First, we note that the metric ansatz implies

$$M = \frac{4}{3}\pi r_+^3 P - \frac{h(r_+, P) r_+}{2}.$$

This then determines V via the first law, and allows us to define a specific volume

$$v = k\frac{V}{N}, \qquad N = \frac{A}{L_{\text{pl}}^2},$$

where $A = 4\pi r_+^2$ is the horizon area, N is interpreted as the number of degrees of freedom associated with the horizon, and $k = 4(d-1)/(d-2) = 6$; the latter is a definition carried over from previous work. Then

$$v = \frac{k}{4\pi r_+^2}\left[\frac{4}{3}\pi r_+^3 - \frac{r_+}{2}\frac{\partial h(r_+, P)}{\partial P}\right],$$

and the black hole temperature is

$$\begin{aligned} T &= \frac{f'(r_+)}{4\pi} \\ &= 2r_+ P - \frac{h(r_+, P)}{4\pi r_+} - \frac{1}{4\pi}\frac{\partial h(r_+, P)}{\partial r_+}. \end{aligned}$$

It remains to match this to the van der Waals equation of state,

$$T = \left(P + \frac{a}{v^2}\right)(v - b).$$

To solve this, the authors use the ansatz

$$h(r, P) = A(r) - PB(r),$$

which reduces the PDE to a system of two ODEs. The authors find a solution for A and B, and substitute these back into the ansatz for h, then finally into the metric function f to obtain

$$f = 2\pi a - \frac{2M}{r} + \frac{r^2}{\ell^2}\left(1 + \frac{3b}{2r}\right) - \frac{3\pi ab^2}{r(2r+3b)} - \frac{4\pi ab}{r}\ln\left(\frac{r}{b} + \frac{3}{2}\right).$$

They note that the terms logarithmic and linear in r are "strange," but otherwise, the requirement for positivity of a (which signifies attractive interactions) implies a spherical topology for the black hole horizon. It is noted that this metric is qualitatively similar to an exact solution to the field equation for a certain class of conformal gravity theories [32]. Finally, the authors examine the stress-energy tensor corresponding to this solution. They find that at large r,

$$\rho = \frac{P}{2} - \frac{Pb}{2r} + \frac{1 - 2\pi a}{16\pi r^2} + \frac{ab}{4r^3} + \cdots,$$

$$p_2 = \frac{Pb}{2r} + \frac{ab}{4r^3} + \cdots,$$

whose corresponding matter content remains to be determined.

- The authors note that this procedure can be used to construct a black hole metric from a given (thermodynamic) equation of state. For instance, one can construct a black hole whose pressure equation of state coincides with the virial expansion of a given equation of state.

- This is interesting, because you've previously identified ordinary systems with nonzero virial coefficients but flat thermodynamic geometry. Could you do the same here to construct black hole solutions with flat thermodynamic metric?

- Delsate and Mann [33] generalize this work to vdW black holes in d dimensions, and argue that the "vdW" black hole metric should be interpreted as a near-horizon metric.

4.2.3 Polytropic black hole

- Notes from Setare and Adami [34]. They use the method outlined previously by Rajagopal et al. [31] to show that there is a static spherically symmetric BH metric that corresponds to a polytropic pressure EOS. The metric is

$$ds^2 = -f(r)\,dt^2 + \frac{dr^2}{f(r)} + r^2\,d\Omega^2,$$

with metric function f of the form

$$f(r) = \frac{r^2}{\ell^2} - \frac{2M}{r},$$

and the corresponding stress-energy tensor is

$$T^\mu_\nu = \frac{1}{8\pi r^2}\,\mathrm{diag}(1, -1, 0, 0),$$

which satisfies all three energy conditions. The temperature of the black hole is

$$T = \frac{f'(r_h)}{4\pi} = \frac{3}{4\pi} \frac{r_h}{\ell^2},$$

where r_h is the horizon radius. This is said to coincide with a polytropic EOS of the form $P = K\rho^{-2}$, where K is a positive constant and ρ is the energy density of the gas. (So their metric does not represent all polytropic gases, only a particular value of γ.)

- They then investigate the corresponding Carnot cycle with this polytropic gas as working substance, which for this solution happens to coincide with the Stirling cycle. Surprisingly, the efficiency for this process turns out to be

  $$\eta = 1 - \frac{T_C}{T_H},$$

  which is the same as the classical Carnot efficiency. There is a statement about how this holds for all static black holes.

4.3 Superfluid black holes

- **(reading list)** Notes from Hennigar et al. [35]. Presents a "λ-line" phase transition in black hole thermodynamics, which is a line of continuous second-order phase transitions. More details are presented in a 2017 JHEP article by the same authors. The idea is to consider the linearized EOM of Lovelock black holes with conformal scalar hair arising from the coupling of a real scalar field to dimensionally extended Euler densities.

5 Optimal finite-time processes

- Notes from Bravetti et al. [22].
- We introduce the quantity
$$A = U - T_0 S + p_0 V,$$
also called the *availability* or exergy of the system. The subscript zero refers to intensities associated with the environment. The change in A from an initial state to a final state for the system equals (minus) the maximum amount of work that can be extracted (reversibly) from a system that is in contact with a thermal reservoir; that is,
$$W^{\max} = -\Delta A = -\Delta U + T_0 \Delta S - p_0 \Delta V.$$
Note that in order for the system to extract work from its environment, it cannot be in thermal equilbrium with it, for then $\Delta A = 0$. Further details can be found in Volume 5 of Landau et al. [36].
- If the process is not performed reversibly, then some of this available work must be lost to dissipation. Call this quantity $(\Delta A)_{\text{dest}}$, the availability lost due to irreversibility. Salamon and Berry [12] showed that the thermodynamic length derived from the Weinhold metric provides a lower bound on this availability loss:
$$(\Delta A)_{\text{dest}} \geq L_U^2 \, \frac{\epsilon}{\tau},$$
where ϵ is a mean relaxation time that depends on the system and τ is the total duration of the process. It is implicitly assumed in the derivation that $\epsilon \ll \tau$.
- Thus, the maximum work that can be extracted from a system in contact with a reservoir, as a function of the duration τ of the process, is given by
$$W^{\max}(\tau) = -\Delta A = -\Delta U + T_0 \Delta S - p_0 \Delta V - L_U^2 \, \frac{\epsilon}{\tau}.$$
This bound is useful because it includes information about the irreversibility without having to assume a particular microscopic model. Note that ordinary thermodynamics merely predicts that $(\Delta A)_{\text{dest}} \geq 0$, and that the finite-time result reduces to this in the limit $\tau \to \infty$, as we expect.
- The bound on $(\Delta A)_{\text{dest}}$ becomes an equality when the thermodynamic speed dL_U/dt of the process is constant. This gives an explicit condition for the best (i.e. minimal availability loss) protocol for a given path. Moreover, for a given pair of initial and final states, the shortest path joining the two (i.e. the one with minimum length L_U) is a geodesic of the metric.
- One can also use the Ruppeiner metric to derive a similar bound. One has
$$(\Delta S)_{\text{irr}} \geq L_S^2 \, \frac{\epsilon}{\tau},$$
where $(\Delta S)_{\text{irr}}$ represents the irreversible entropy production of the system and environment during the process.
- The idea of [22] is to bring this concept over to black hole thermodynamics. Proceeding by analogy with ordinary thermodynamics, the maximum amount of work that can be extracted (reversibly) from a Kerr black hole immersed in an environment is
$$W_{\max}^{\text{non-isol}}(\tau) = -\Delta M + T_0 \Delta S + \Omega_0 \Delta J - L_U^2 \, \frac{\epsilon}{\tau},$$

where ϵ and τ are as before, and T_0 and Ω_0 are temperatures and angular velocities associated with the environment. Other black hole solutions (e.g. RN, Kerr-AdS) will have additional intensive/extensive variables that need to be accounted for here, but for now we will just look at Kerr. As we'll see shortly, we will run into a couple of issues, both of which stem from the fact that the Weinhold metric for Kerr is not positive-definite.

- It's interesting that this concept of availability requires an environment. It seems like it should be possible to say something about dissipation for isolated systems, too.

5.1 Optimal Penrose processes for Kerr black holes

- The authors of [22] have a rather loose definition of Penrose process; they mean any process by which the angular momentum of a black hole is brought to zero. They study initially extremal black holes which evolve to Schwarzschild black holes of unspecified final mass:

$$(M_i,\ J_i = M_i^2) \quad \to \quad (M_f,\ 0).$$

In this way, there are two unknowns – the final black hole, which is specified by the final mass M_f, and the state-space path connecting the initial and final black holes.

- Since the view of the paper is to regard geodesics of the Weinhold metric as processes of special physical significance (i.e. minimum-dissipation processes), they move to the coordinates

$$M(t,x) = \frac{t^2 - x^2}{4}, \qquad J(t,x) = \frac{tx(t^2-x^2)^3}{4(t^2+x^2)^2}.$$

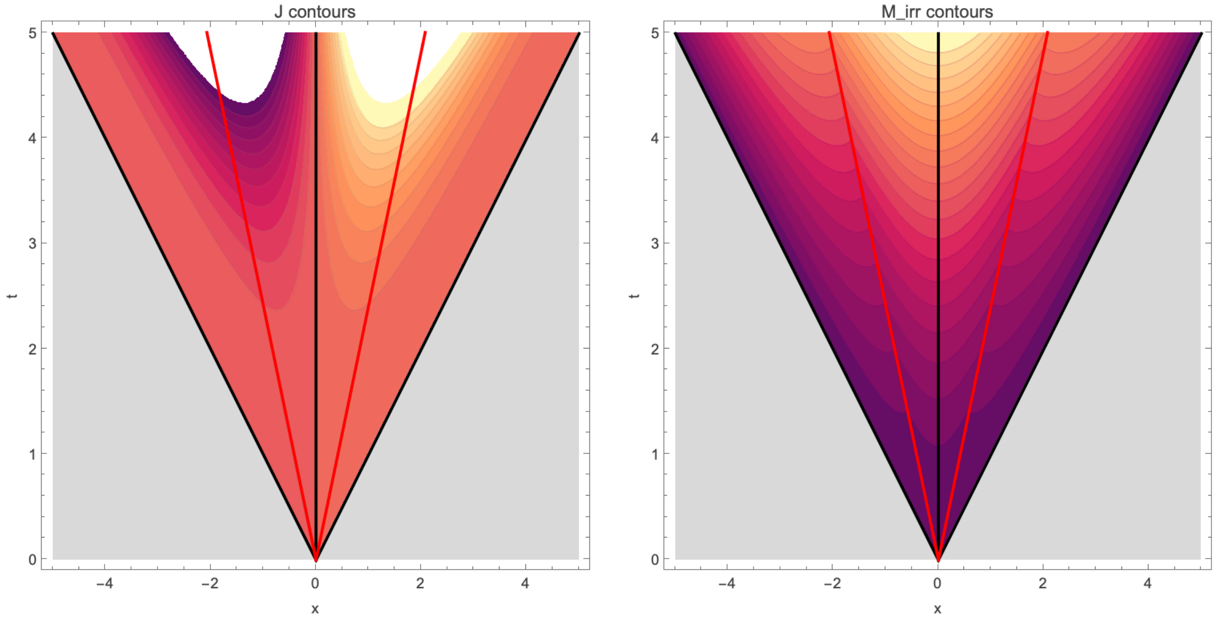
[This corrects Eq. (13) in the paper, which gives the wrong denominator for $J(t,x)$.] In these coordinates, the Weinhold metric takes the form of Minkowski:

$$ds^2 = -dt^2 + dx^2.$$

This is the motivation for the coordinate transformation: the (x,t) coordinates make the Weinhold geometry more transparent. Processes in (x,t) space can be understood using our intuition about the Minkowski metric; for example, geodesics in these coordinates are straight lines. The same coordinates were used by Åman et al. [37] to study the thermodynamic geometry of the Kerr solution. It is important to note that x and t are constrained to lie above a pair of lines, corresponding to extremal black holes, in the (x,t) state space.

- In these coordinates, the entropy is

$$S(t,x) = \frac{(t^2-x^2)^4}{4(t^2+x^2)^2}.$$



- Extremal curves are plotted as red lines in the figure. Aman et al. consider also the region under this, but this corresponds to the inner horizon of the Kerr black hole, which has a different entropy function and temperature. We will restrict to the region above the red curves; this gives an onto, one-to-one mapping to the full physical (i.e. sub-extremal) (M, J) space.

- From the definitions, one can see that all black holes with zero spin and nonzero mass lie on the curve $x = 0$. Thus, the initial Kerr black hole lives on the extremal red curve, and the final Schwarzschild black hole lives on the vertical black curve.

- Then the authors of [22] give the following arguments: (1) geodesics of the Weinhold metric are length-minimizing; (2) consider the set of all (physically admissible) paths joining an initially extremal black hole to a final Schwarzschild black hole. The lengths of these paths are bounded below; there exists a path with minimal length. By (1), this path must be a geodesic. Then since thermodynamic length is proportional to dissipation, this geodesic represents a minimum-dissipation process.

- There are an infinite number of paths joining two points in state space. The authors constrain these by requiring the following:

(1) We must have $|dx/dt| > 1$ throughout the process, for thermodynamic stability. Using the Minkowski analogy, the process must be a spacelike curve.

(2) The black hole temperature T must be positive at all times.

(3) The process must obey Hawking's area law theorem. Equivalently, the irreducible mass M_{irr} must never decrease.

The first constraint is unusual – this is the first I've heard of thermodynamic length revealing something about the stability of a process. The method I'm familiar with to check stability is to look at a point in state space, then examine the signs of the second derivatives of your potential. This is a local condition, and is independent of the notion of length. Bravetti's constraint seems like an *ad hoc* requirement to avoid having to consider paths which change character midway. The second constraint is essentially contentless; the entire region above the extremal red curves has positive temperature, so this is always trivially satisfied. One can show this from the definition,

$$T(t, x) = \frac{(t^2 - x^2 - 2tx)(t^2 - x^2 + 2tx)}{2(t^2 - x^2)^3},$$

together with the fact that the extremal curves satisfy

$$t = \pm\sqrt{\frac{1+\sqrt{2}}{-1+\sqrt{2}}}\, x.$$

For later, we propose an additional constraint, which was not considered by [22].

(4) Since this path is meant to represent a Penrose process, we should have $dJ \leq 0$ throughout.

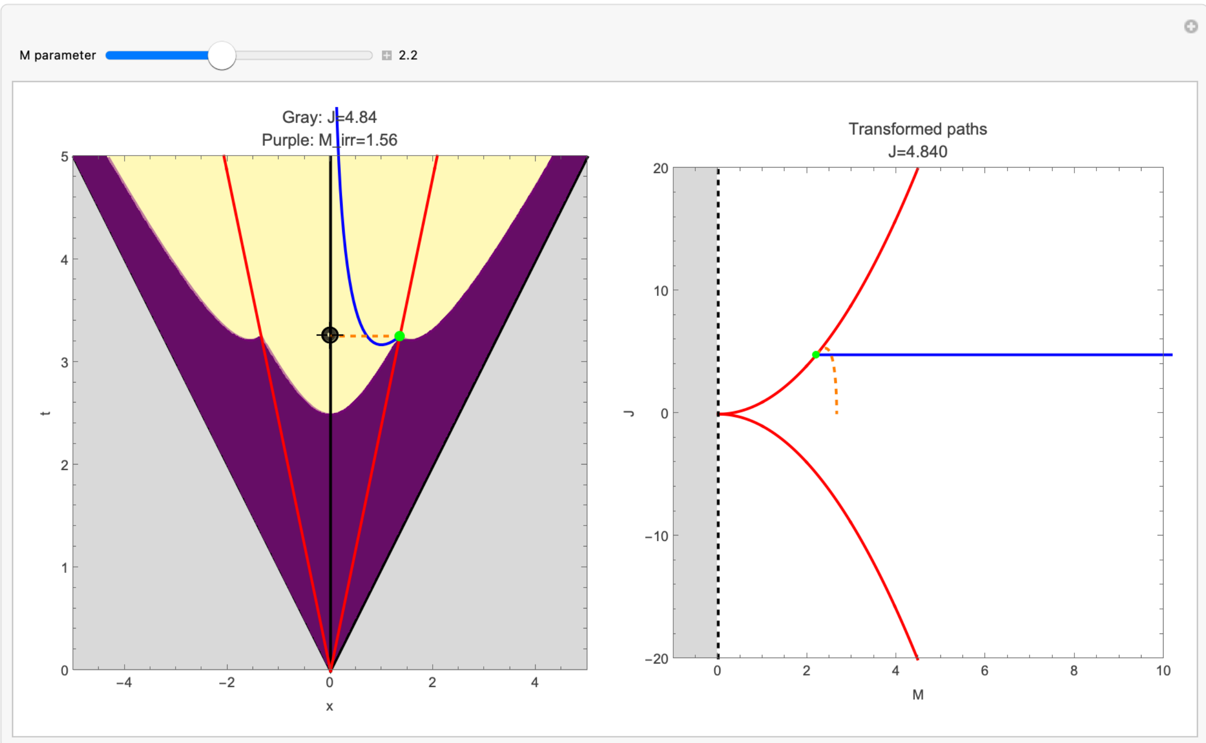
- Using constraints (1)–(3), the authors identify the following process to be extremal. For a given initial black hole mass M_i, the black hole spin evolves as

$$J(M) = \frac{\sqrt{2M_i}\sqrt{1+\sqrt{2}}\sqrt{2M_i\sqrt{2}+2M_i-4M}}{(M_i\sqrt{2}+M_i-M)^2}M^3,$$

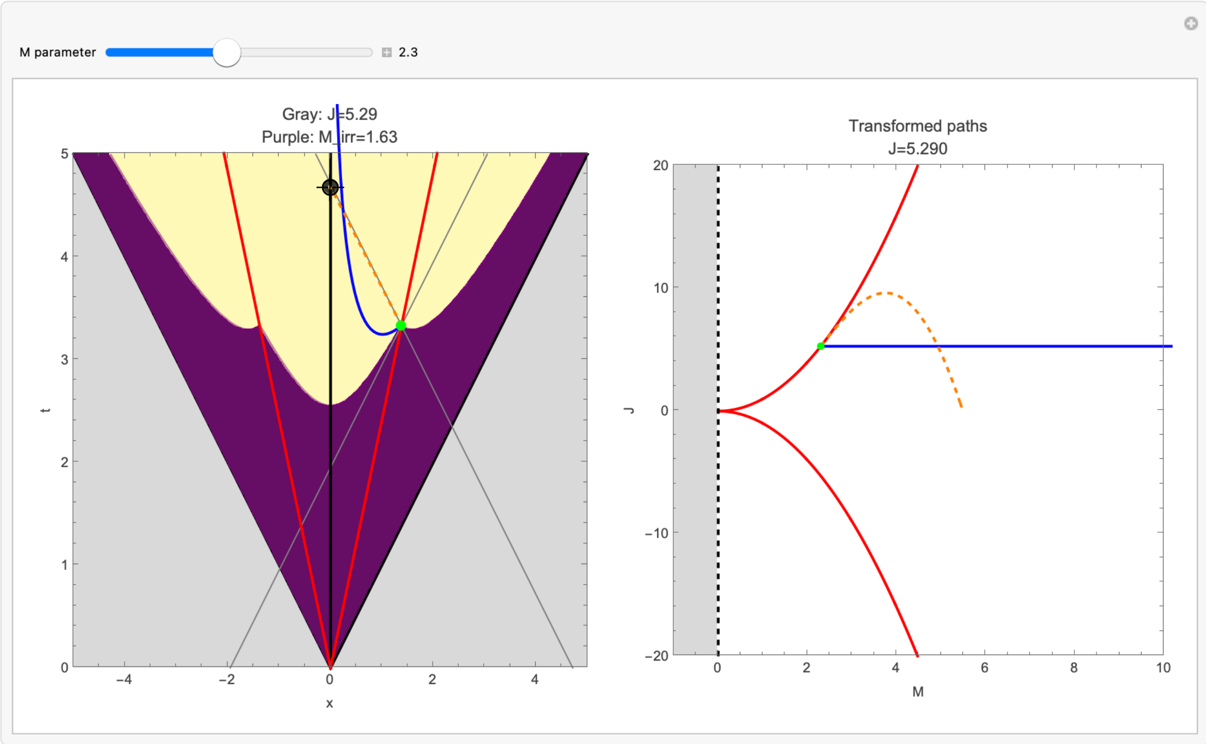
with corresponding thermodynamic length

$$L(M_i) = \sqrt{2M_i(\sqrt{2}-1)}.$$

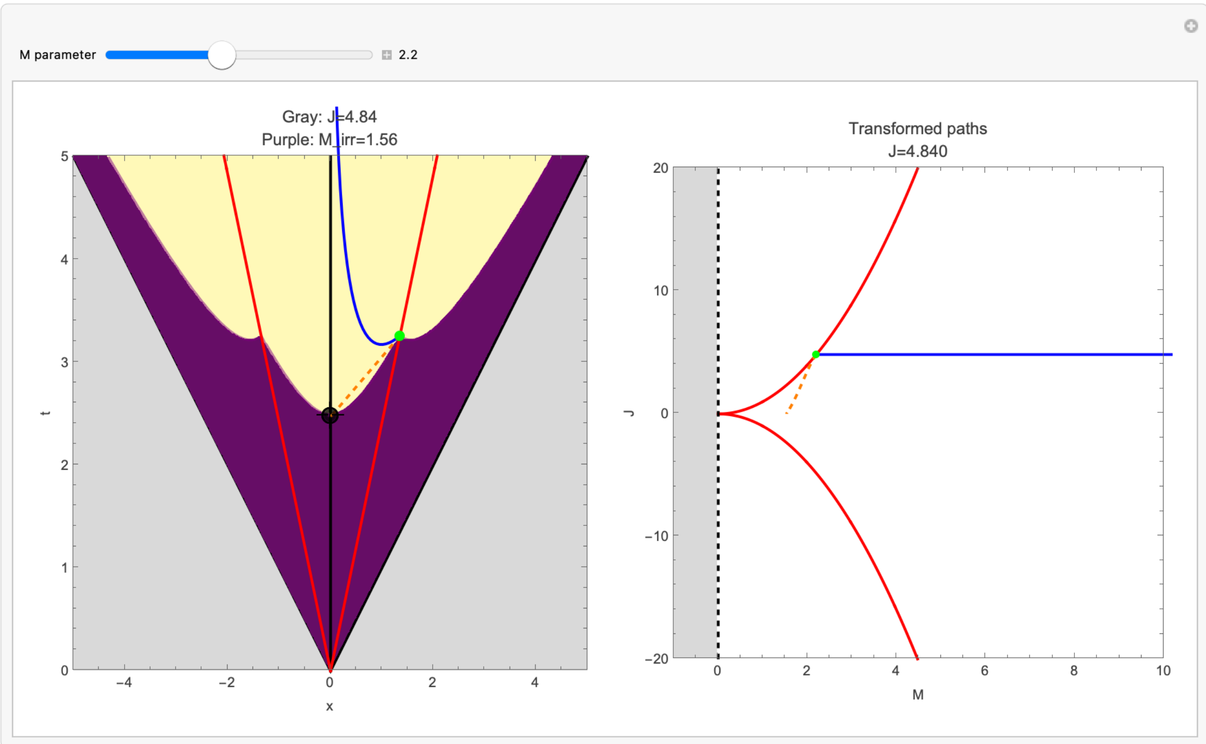
This is plotted as a dashed orange curve in the physical (M,J) space on the right. The corresponding picture in (x,t) space is a horizontal line joining the extremal curve to the zero-spin curve. Shown in blue is the contour of J corresponding to the initial spin, and the yellow-purple coloring marks a contour of M_{irr} corresponding to the initial irreducible mass. Notice that J increases before it starts to decrease, in violation of our proposed constraint (4).



This is the main result of the paper; the authors claim that this process minimizes dissipation. But it is easy to draw a process with zero length; we give an example below.



This process also satisfies constraints (1)–(3), so by the authors' criteria, this is a valid physical process which corresponds to zero dissipation. We can be a bit stricter, and require that our process be more Penrose-like by imposing constraint (4). One example is the following:



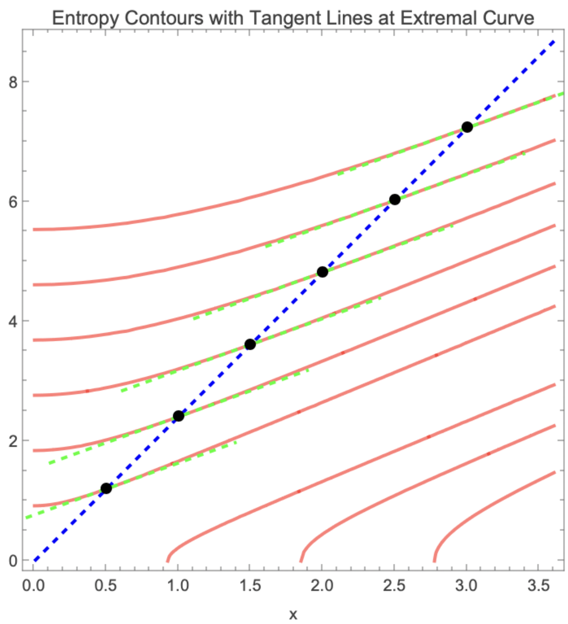
Strictly speaking, the curve drawn violates (3); if we want to do this, we need to ensure that the curve crosses all contours of M_{irr} in the increasing direction.

- In any case, the Lorentzian signature of the metric makes things subtler. One can always draw a smooth spacelike path with length arbitrarily close to zero joining a pair of spacelike-separated points; just zig-zag along null curves, then round the corners. One can also draw a spacelike path of arbitrarily large length joining any two spacelike-separated points, if you drop the requirement of smoothness; just zig-zag to add more horizontal segments.

- Many statements about geodesics in general relativity (e.g. their length-minimizing character) require causal geodesics or smoothness, neither of which are natural ideas in thermodynamic geometry. It seems as though the statement $dM_{\text{irr}} \geq 0$ will play the role of causality, as far as constraining the form of physical paths is concerned.

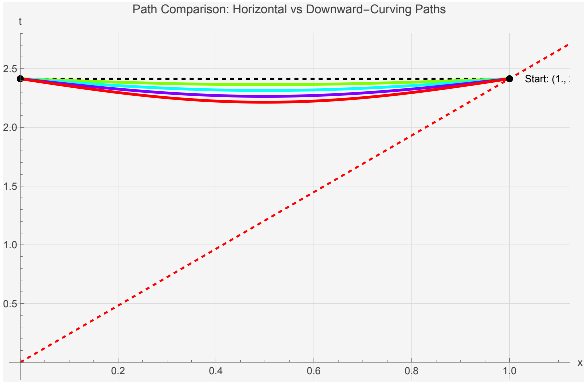
- Notice, for example, that it isn't possible to join an extremal black hole to a Schwarzschild black hole via a zero-length, zig-zagged path if we impose both (3) and (4). We can see this in the second figure, by considering the very start of the process (the green solid point). Travelling in the null direction upward corresponds to $dJ > 0$, while the downward null direction corresponds to $dM_{\text{irr}} < 0$. It also isn't clear whether there exists a geodesic connecting an extremal black hole to a Schwarzschild black hole which obeys (3) and (4) at all points.

- One can show that the tangent to the curves of constant entropy, at the location of the extremal curve, has constant slope of around 0.867. This implies (together with visual inspection of the entropy contours) that a constant-entropy process is everywhere spacelike, which is reassuring.

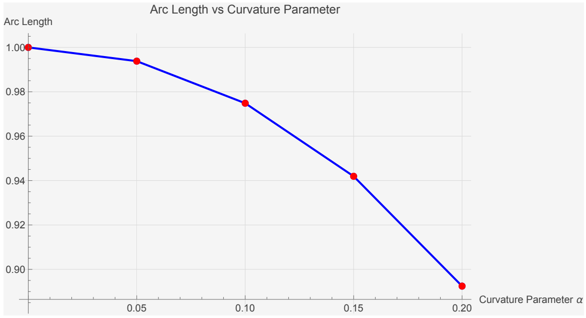


- On the other hand, it appears as though the process identified in [22] does not minimize length between the initial and final points. One can construct downward-curving paths which respect the entropy-increase law but have smaller length than a horizontal line. These are also smooth, and everywhere spacelike, so they satisfy criteria (1)–(3). We summarize our findings below.

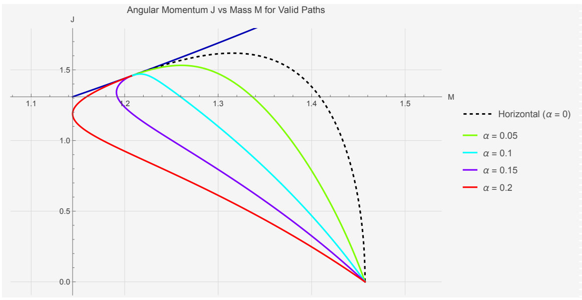
- In summary, it's not really clear what Bravetti's results mean. We have shown that there exist physical paths of arbitrarily small dissipation, by travelling in the upward null direction. We have also shown that there exist smooth spacelike paths, obeying criteria (1)–(3) and joining two points with purely horizontal separation, which have shorter length than a geodesic (i.e. horizontal line). As we mentioned at the start, the problem has to do with the fact that the metric is not positive-definite.



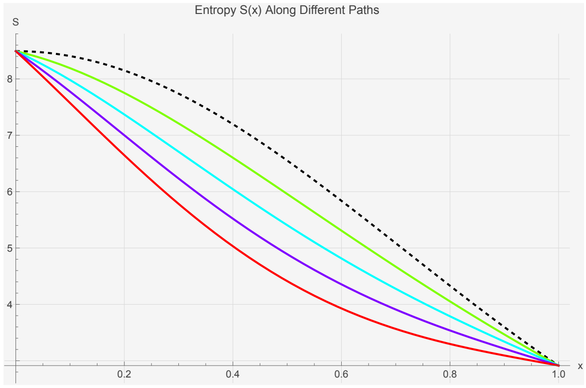
(a) Downward-curving paths in (x, t) space are shown as solid colored lines. Each curve is spacelike.



(b) Length of each curve as a function of the curvature parameter α, which is just a factor multiplying a sine function.



(c) The same paths in (M, J) space. Note that some of these paths (specifically, the ones with smaller length) correspond to processes where both M and J are initially decreasing.



(d) Entropies along these paths, parametrized by x. Since x decreases along each path as one moves from an extremal black hole to a Schwarzschild one, these should be read from right to left.

Figure 7: Comparison of different paths in thermodynamic state space; this shows that the process of Bravetti et al. [22] is not the optimal path joining their initial and final black holes.

5.2 Open question: overcharging and charged Penrose processes

- Note that the Reissner-Nordström black hole has a flat Ruppeiner metric; therefore, it may be possible to repeat the method of Bravetti for an overcharging process. The main difference is that the geodesics will be entropy-minimizing. This may be used to, for example, provide a lower bound on entropy production for a given process time τ.
- On a related note, it is worth studying charged Penrose processes, because you will encounter the same quantities for S and T that you'd study in an overcharging scenario.

5.3 Digression: Optimal control in nonequilibrium systems

- These papers are nice, since they represent what we want to do for black-hole systems. The main difference is that the metrics considered here are Riemannian, which implies that geodesics are length-minimizing.
- Sivak and Crooks [38], motivated by the study of molecular-scale machines, consider the role of thermodynamic geometry in determining optimal paths. One complication in the study of these machines is that they typically operate out of equilibrium. For instance, one might

have a rotary $F_O F_1$–ATPase motor powered by proton flow across a gradient ($\sim$ 200 MeV free energy per proton). This contribution dwarts the characteristic energy scale of thermal fluctuations, $k_B T \approx 25$ meV. Another feature of interest is efficiency. Broadly speaking, we want to be able to abstract away the molecular details.

- Macroscopic processes: investigate optimal processes using thermodynamic length. It is claimed that the metrics of Weinhold, Ruppeiner, and Schlögl, based on different thermodynamic potentials, are all essentially equivalent in the thermodynamic limit. (This is false.)

- Another important result was derived by assuming endoreversibility, where the system and surroundings are in thermal equilibrium at all times, but not necessarily with each other. For a system driven by a single control parameter, this corresponds to the system trailing the environment at any instant, residing in equilibrium corresponding to the system at the previous value of the control parameter.

- Crooks has a couple of papers [39, 40] which formulate a microscopic version of thermodynamic length. The aim of the present work is to derive a thermodynamic length for microscopic systems without the assumption of endoreversibility. With this metric:
  - Optimal paths (which minimize dissipation) are geodesics.
  - Dissipation is inversely proportional to protocol duration.
  - The optimal control parameter path is independent of duration.
  - Optimal protocols perturb the control parameter so as to accumulate excess work at a constant rate.

- In this work, the physical system is in contact with a thermal bath. The probability distribution over microstates x at equilibrium is given by the canonical ensemble:

$$\pi(x|\boldsymbol{\lambda}) = \exp[\beta(F(\boldsymbol{\lambda}) - E(x, \boldsymbol{\lambda}))],$$

  where F is the free energy, E is the system energy, and $\boldsymbol{\lambda}$ are experimentally controllable system parameters. For example, a control parameter for a gas in a cylinder could be the position of the piston imposing a particular accessible volume.

- In equilibrium, the macroscopic state of the system (i.e. probability distribution over microstates) is completely specified by $\boldsymbol{\lambda}$. But out of equilibrium, the probability distribution depends on the history of $\boldsymbol{\lambda}$. We denote this by the control parameter protocol $\boldsymbol{\Lambda}$. Assume that this is smooth enough to be twice time-differentiable.

- As another digression, there is an interesting citation of a study of the time-dependent potential

$$U(x, t) = \frac{k}{2}(x - ut)^2,$$

  which represents a moving potential well. One picture: the particle is attached to a spring, the other end of which moves at a constant speed u. Thermal forces are modeled as a sum of linear friction and white noise, and the motion is assumed overdamped. You get

$$\dot{x} = -\frac{k}{\lambda}(x - ut) + \tilde{V}(t),$$

  where γ is the coefficient of friction, and $\tilde{V}(t)$ is delta-correlated white noise with variance $2/\beta\gamma$,

$$\langle \tilde{V}(t_1)\tilde{V}(t_2)\rangle = \frac{2}{\beta\gamma}\,\delta(t_2 - t_1)$$

  as mandated by the fluctuation-dissipation theorem.

- Back to Sivak and Crooks: the average instantaneous rate of energy flow into the system is $d\langle E\rangle_{\mathbf{\Lambda}}/dt$. This is the sum of average instantaneous rate of heat flow Q from the environment into the system,
$$\frac{d\langle Q\rangle_{\mathbf{\Lambda}}}{dt} \equiv \left\langle \frac{dx}{dt}^T \frac{dE}{dx} \right\rangle_{\mathbf{\Lambda}}$$
and the average instantaneous power P due to external perturbations of the control params $\boldsymbol{\lambda}$,
$$P \equiv \frac{d\langle W\rangle_{\mathbf{\Lambda}}}{dt} = -\frac{d\boldsymbol{\lambda}}{dt}^T \cdot \langle \mathbf{X}\rangle_{\mathbf{\Lambda}},$$
where $\mathbf{X} \equiv -\partial E/\partial\boldsymbol{\lambda}$. The angle brackets denote nonequilibrium averages over the ensemble, following the protocol $\mathbf{\Lambda}$.

- The average power of a system initially at equilibrium at time t_0 and control parameters $\boldsymbol{\lambda}(t_0)$, average heat flow vanishes **(clarify)** and
$$P(t_0) = -\left[\frac{d\boldsymbol{\lambda}}{dt}\right]_{t_0}^T \cdot \langle \mathbf{X}\rangle_{\boldsymbol{\lambda}(t_0)},$$
where angle brackets denote an ensemble average at these fixed control parameters. Thus for an arbitrary protocol $\mathbf{\Lambda}$ with current parameter values $\boldsymbol{\lambda}(t_0)$, the excess power compared to equilibrium is
$$P_{\text{ex}}(t_0) = -\left[\frac{d\boldsymbol{\lambda}}{dt}\right]_{t_0}^T \cdot \langle \Delta\mathbf{X}(t_0)\rangle_{\mathbf{\Lambda}},$$
where $\Delta\mathbf{X}(t_0) \equiv \mathbf{X}(t_0) - \langle\mathbf{X}\rangle_{\boldsymbol{\lambda}(t_0)}$. Dynamic linear-response theory expresses this deviation, to first order in the magnitude of the external perturbation, by
$$\langle \Delta\mathbf{X}(t_0)\rangle_{\mathbf{\Lambda}} \approx \int_{-\infty}^{t_0} dt'\, \boldsymbol{\chi}(t_0 - t') \cdot [\boldsymbol{\lambda}(t') - \boldsymbol{\lambda}(t_0)],$$
where
$$\chi_{ij}(t) \equiv \beta\, \frac{d\Sigma_{ij}^{(\boldsymbol{\lambda}(t_0))}}{dt}$$
is the response of conjugate force X_j at time t to a perturbation in the control parameter λ^i at time zero, and
$$\Sigma_{ij}^{(\boldsymbol{\lambda}(t_0))}(t) \equiv \langle \delta X_j(0)\delta X_i(t)\rangle_{\boldsymbol{\lambda}(t_0)}$$
is the coefficient of the covariance matrix for conjugate forces X_j and X_i separated by time t at constant control parameters $\boldsymbol{\lambda}(t_0)$.

- Assumptions: over time scales where the response function $d\mathbf{\Sigma}^{(\boldsymbol{\lambda}(t_0))}/dt\,|_\tau$ is significantly different from zero, both $\langle\Delta\mathbf{X}(t_0)\rangle_{\mathbf{\Lambda}}$ and $\langle\mathbf{X}\rangle_{\boldsymbol{\lambda}(t_0)} - \langle\mathbf{X}\rangle_{\boldsymbol{\lambda}(t_0-\tau)}$ are linear in $\boldsymbol{\lambda}(t_0) - \boldsymbol{\lambda}(t_0 - \tau)$. Assume further that control parameter velocities change on time scales slower than the relaxation time of the system's force flucutations. Then
$$P_{\text{ex}}(t_0) = \left[\frac{d\boldsymbol{\lambda}}{dt}\right]_{t_0}^T \cdot \beta \int_0^\infty dt''\, \mathbf{\Sigma}^{(\boldsymbol{\lambda}(t_0))}(t'') \cdot \left[\frac{d\boldsymbol{\lambda}}{dt}\right]_{t_0}$$
In principle you include the time-dependent covariance matrix, but you get a local expression
$$P_{\text{ex}}(t_0) = \left[\frac{d\boldsymbol{\lambda}}{dt}\right]_{t_0}^T \cdot \boldsymbol{\zeta}(\boldsymbol{\lambda}(t_0)) \cdot \left[\frac{d\boldsymbol{\lambda}}{dt}\right]_{t_0}.$$
You then integrate this over the control parameter protocol to get the mean excess work:
$$W_{\text{ex}} = \int_0^{\Delta t} dt\, P_{\text{ex}}(t) \sim \left|\frac{d\boldsymbol{\lambda}}{dt}\right|.$$

- Now for the important points: any covariance matrix is symmetric and positive semidefinite. Moreover, the conjugate-force covariance matrix $\mathbf{\Sigma}^{(\boldsymbol{\lambda}(t_0))}(t)$ varies smoothly with control parameter values, except at macroscopic phase transitions. An important caveat is that this assumes $\mathbf{\Lambda}$ is smooth in time. By construction, this excludes optimal paths with jumps in $\boldsymbol{\lambda}$, which do exist.

- Thus the friction tensor $\boldsymbol{\zeta}$ induces a Riemannian manifold on the space of thermodynamic states. Positive semidefiniteness ensures that excess power and work are non-negative, which is consistent with the second law.

- The notions of generalized thermodynamic length

$$\mathcal{L} = \int_0^{\Delta t} dt\,\sqrt{P_{\text{ex}}(t)}$$

and divergence

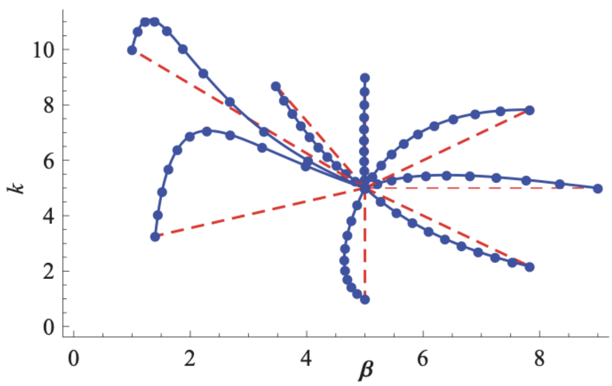
$$\mathcal{J} = \Delta t \int_0^{\Delta t} dt\, P_{\text{ex}}(t)$$

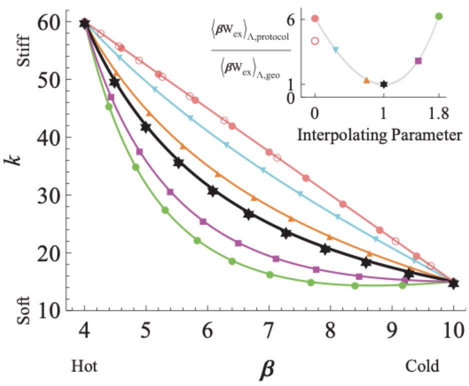
then arise naturally. Notice that $\mathcal{J} = \Delta t\, W_{\text{ex}}$. For a fixed control parameter path, the corresponding thermodynamic length is independent of the time interval Δt and the relative control parameter velocities en route. This places a lower bound on the excess work:

$$W_{\text{ex}} \geq \frac{\mathcal{L}^2}{\Delta t}.$$

Optimal paths have constant $|d\boldsymbol{\lambda}/dt|$.

- **(reading list)** Zulkowski et al. [41] consider the Riemannian metric induced on the space of controllable parameters describing a system driven out of equilbrium, in the same thread as the previous paper. An important modification is that the temperature may change during the protocol. In this framework, optimal protocols, which minimize dissipation, are geodesics of the metric. They find analytic expressions for these geodesics, and show numerically (by solving the Fokker-Planck equation) that these protocols do indeed minimize dissipation. The geometry is surprisingly rich; they solve their equations by finding Killing vectors of the metric and making use of the corresponding conserved quantities.





The dotted blue paths represent geodesics in the (β, k) parameter space; the dashed red lines represent naive protocols that interpolate linearly between the start and end-points. The inset in the plot at right shows that the geodesic path (in black) corresponds to the smallest dissipation. The other colors represent paths that interpolate between the linear path and the geodesic.

- **(reading list)** Rotskoff and Crooks [42] apply these ideas to the Ising model, in order to study optimal protocols for reversing the net magnetization. Since the metric in this case is not expressible in analytic form, they compute the metric numerically.

6 Heat engines

We discuss a few results on heat engines in ordinary and finite-time thermodynamics.

6.1 Efficiency of a Carnot engine

- The efficiency of a Carnot engine operating between a hot reservoir at temperature T_H and cold reservoir at T_C is given by
$$\eta_C = 1 - \frac{T_C}{T_H}.$$
All reversible engines have the same efficiency; furthermore, this is the maximum possible efficiency of any engine. For a proof, see, e.g., Vol. 1 of the Feynman lectures.

- A Carnot cycle operates in four stages: (1) an isothermal expansion in contact with a hot reservoir, (2) an adiabatic expansion to lower the temperature to that of the cold reservoir, (3) isothermal compression in contact with the cold reservoir, (4) adiabatic compression to return the system to the temperature of the hot reservoir. The ideal Carnot cycle is not practical, because the isothermal heating and cooling stages take an infinitely long time to complete; therefore, the power output $P = dW/dt$ is zero. This is because the processes are designed to be *reversible*, so the temperature of the working substance can differ only infinitesimally from the reservoir it is in contact with. This requirement is dropped in the next section.

6.2 Curzon and Ahlborn

- An early paper (the first?) in finite-time thermodynamics [43]. The idea is to consider a Carnot-like cycle which is capable of outputting nonzero power. Thus there are isothermal and adiabatic compression/expansion stages, but now we allow the reservoir and working-substance temperatures to differ. This is necessary to get nonzero heat flux, and therefore nonzero power.

- The first assumption is the that heat flux Φ_1 during the isothermal expansion is constant, so
$$\Phi_1 = \alpha(T_1 - T_{1w}),$$
where α is a constant dependent on the thickness of thermal conductivity of the wall, T_1 is the temperature of the heat source, and T_{1w} is the temperature of the working substance during this process. Then the energy input over a time t_1 is
$$W_1 = \alpha t_1(T_1 - T_{1w}).$$
Similarly, during the isothermal compression, heat is rejected
$$W_2 = \beta t_2(T_{2w} - T_2).$$
It is assumed that the adiabatic expansion and compression are reversible, so that
$$\frac{W_1}{T_{1w}} = \frac{W_2}{T_{2w}}.$$
The final assumption is that the total time of the cycle is $(t_1 + t_2)\gamma$, and
$$(t_1 + t_2)\gamma \;=\; \underbrace{(\gamma - 1)(t_1 + t_2)}_{\text{Adiabatic processes}} \;+\; \underbrace{(t_1 + t_2)}_{\text{Isothermal processes}}\;.$$
Thus, if the engine is sped up by some factor, the total times taken for both types of process increase by the same factor.

- With these assumptions, the power output of the engine is

$$P = \frac{W_1 - W_2}{(t_1 + t_2)\gamma},$$

and the previous considerations allow us to determine the temperatures T_{1w} and T_{2w} of the working substance that maximize power output. The answer is

$$T_{1w} = CT_1^{1/2}, \qquad T_{2w} = CT_2^{1/2},$$

where C is a constant given by

$$C = \frac{(\alpha T_1)^{1/2} + (\beta T_2)^{1/2}}{\alpha^{1/2} + \beta^{1/2}}.$$

One then finds that the maximum power output is

$$P_{\max} = \frac{\alpha\beta}{\gamma}\left(\frac{T_1^{1/2} - T_2^{1/2}}{\alpha^{1/2} + \beta^{1/2}}\right)^2.$$

- The main result of the paper is the efficiency η corresponding to this maximum power. They show that

$$\eta = 1 - \sqrt{\frac{T_2}{T_1}}.$$

Like the Carnot engine, the result depends only on the temperatures of the hot and cold reservoirs. However, this efficiency is smaller than the Carnot efficiency, as expected.

- Here are the main assumptions: (1) the times taken for the adiabatic and isothermal processes are directly proportional, (2) the ingoing and outgoing heat fluxes are constant, (3) we can perform our reversible processes (adiabatic expansion and compression) in finite time.

- A brief summary: in order to get nonzero power output, we need to consider finite temperature differences. But this takes us away from reversibility, and therefore the efficiency is lower than the Carnot bound. What's interesting is that, numerically, the efficiency at maximum power output seems to agree with observations of real engines. This is consistent with the idea that real engines are designed in a way that maximizes power output.

6.3 Responses to Curzon-Ahlborn

- Van den Broeck [44] shows that "the Curzon-Ahlborn efficiency is a fundamental result that follows, without approximation, from the theory of linear irreversible thermodynamics." This is a strong statement about heat engines; may be worth reading the argument in detail.

- On the other hand, Lavenda [45] criticizes the CA derivation, arguing that there is an unexamined assumption about how long it takes for the parts of the system to achieve thermal equilibrium with each other. Because the time for it takes for this "adiabatic equilibriation" to occur cannot be constrained by thermodynamics, it follows that the total time of the process cannot be determined either. Instead, he argues that the CA efficiency is a result that follows from the principle of maximum work: the final mean temperature at which adiabatic equilibriation occurs yields the maximum work. This then yields a maximum efficiency, which Lavenda shows is the same as the CA efficiency. It is emphasized that in this derivation, there are no assumptions made about the rates of processes. Indeed, "work done" is something you can specify without reference to time.

- Lavenda claims that CA's answer is correct, but their methods are invalid. That is, it is mere coincidence that in their model, the temperature that maximizes their definition of power output happens to be the same temperature which yields maximum extractable work, and therefore maximum efficiency.

- Schmiedl and Seifert [46] study a class of cyclic Brownian heat engines using finite-time thermodynamics. These engines operate at the Carnot efficiency for infinitely long cycle times, but when operating at maximum power, have the efficiency

$$\eta = \frac{2(T_{\rm h} - T_{\rm c})}{3T_{\rm h} + T_{\rm c}},$$

which differs from the CA efficiency. For large reservoir temperature differences $T_{\rm h} - T_{\rm c} \gg T_{\rm c}$, this has the limiting value $\eta \to 2/3$. (The corresponding limit for the Carnot efficiency is 1.) For small differences $\tau \equiv (T_{\rm h} - T_{\rm c})/T_{\rm c} \ll 1$, this becomes

$$\eta = \frac{\tau}{2} - \frac{3\tau^2}{8} + \frac{9\tau^3}{32} + O(\tau^4),$$

which can be compared against the CA efficiency in the same limit,

$$\eta_{\rm CA} = \frac{\tau}{2} - \frac{3\tau^2}{8} + \frac{10\tau^3}{32} + O(\tau^4),$$

which shows that the deviation occurs at third order in τ. This is consistent; the CA efficiency is known to be the efficiency at maximum power output in the linear response regime. Working to higher order, we see that this efficiency is slightly smaller than the CA value. Finally, note that the Carnot efficiency $\eta_C \gg \tau/2$ in the same limit.

- Their main result is to derive a universal law for the efficiency at maximum power of a general class of stochastic heat engines. They find that for these engines,

$$\eta(\alpha) = \frac{\eta_C}{2 - \alpha\eta_C},$$

where η_C is the Carnot efficiency and α depends on an "irreversible action" which can be calculated in principle by solving an optimization problem for the specific engine being considered. In the linear response regime $\tau \ll 1$, this has the limiting value $\eta \to \tau/2$, in agreement with the previous results. When $\alpha = \alpha_{\rm CA}$, this reduces to the CA efficiency. Curiously, when $1 > \alpha > \alpha_{\rm CA}$, this efficiency actually exceeds the CA bound. This is likely because the original CA machine is assumed to be driven by linear Fourier heat flow, but for systems driven strongly out of equilibrium (such as these), nonlinearities can arise. Thus, it is not immediately clear that $\eta > \eta_{\rm CA}$ leads to any inconsistency.

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